

Programming Assignment 1 – TSAM

Question 1(a)

The open TCP ports were:

22/tcp – SSH

2000/tcp – Cisco SCCP

5060/tcp – SIP

```
evie@evies-MacBook-Air ~ % sudo nmap -sS 130.208.246.98
Password:
Starting Nmap 7.991 ( https://nmap.org ) at 2026-08-30 15:23 +0000
Nmap scan report for 130.208.246.98
Host is up (0.0068s latency).
Not shown: 880 filtered tcp ports (no-response), 117 closed tcp ports (reset)
PORT      STATE SERVICE
22/tcp    open  ssh
2000/tcp  open  cisco-sccp
5060/tcp  open  sip

Nmap done: 1 IP address (1 host up) scanned in 4.21 seconds
evie@evies-MacBook-Air ~ %
```

```
evie@evies-MacBook-Air ~ % ping 130.208.246.98
PING 130.208.246.98 (130.208.246.98): 56 data bytes
64 bytes from 130.208.246.98: icmp_seq=0 ttl=50 time=8.936 ms
64 bytes from 130.208.246.98: icmp_seq=1 ttl=50 time=9.273 ms
64 bytes from 130.208.246.98: icmp_seq=2 ttl=50 time=9.549 ms
64 bytes from 130.208.246.98: icmp_seq=3 ttl=50 time=9.560 ms
64 bytes from 130.208.246.98: icmp_seq=4 ttl=50 time=4.622 ms
64 bytes from 130.208.246.98: icmp_seq=5 ttl=50 time=4.586 ms
64 bytes from 130.208.246.98: icmp_seq=6 ttl=50 time=5.490 ms
64 bytes from 130.208.246.98: icmp_seq=7 ttl=50 time=9.274 ms
^C
--- 130.208.246.98 ping statistics ---
8 packets transmitted, 8 packets received, 0.0% packet loss
round-trip min/avg/max/stddev = 4.586/7.661/9.560/2.162 ms
evie@evies-MacBook-Air ~ %
```

Question 1(b)

```
tcpdump: listening on en0, link-type EN10MB (Ethernet), snapshot length 524288 bytes
Remember to check for network connectivity
15:43:11.118241 IP (tos 0x0, ttl 64, id 0, offset 0, flags [DF], proto TCP (6), length 64)
  192.168.1.137.49466 > 130.208.246.98.complex-main: Flags [SEW], cksum 0x1d84 (correct), seq 560349985, win 65535, options [mss 1460,nop,wscale 6,nop,nop,TS val 867987533 ecr 0,sackOK,eol], length 0
  0x0000: 4500 0804 0000 4000 4005 f153 c0a8 0189  E..@.@.S....
  0x0010: 8200 f662 c13a 1388 2166 4321 0000 0000  ...b...fFC1...
  0x0020: 00c2 ffff 1d84 0000 0204 0504 0103 0300  .....
  0x0030: 0101 080a 33bc 704d 0000 0000 0402 0000  ....3.PM.....
15:43:11.122936 IP (tos 0x0, ttl 50, id 0, offset 0, flags [DF], proto TCP (6), length 60)
  130.208.246.98.complex-main > 192.168.1.137.49466: Flags [S.E], cksum 0xbfdd (correct), seq 143520767, ack 560349986, win 65160, options [mss 1412,sackOK,TS val 1722224012 ecr 867987533,nop,wscale 7], length 0
  0x0000: 4500 083c 0000 4000 3206 0d08 8208 f662  E...@.2..X..b
  0x0010: c0a8 0189 1388 c13a 088d f31f 2166 4322  .....fFC"....
  0x0020: a052 fe88 bfdd 0000 0204 0504 0402 080a  .R.....
  0x0030: 66a7 008c 33bc 704d 0103 0307  f...3.PM....
15:43:11.122839 IP (tos 0x0, ttl 64, id 0, offset 0, flags [DF], proto TCP (6), length 52)
  192.168.1.137.49466 > 130.208.246.98.complex-main: Flags [I], cksum 0xe534 (correct), ack 1, win 2057, options [nop,nop,TS val 867987538 ecr 1722224012], length 0
  0x0000: 4500 0834 0000 4000 4006 ffff c0a8 0189  E..4..@.@.....
  0x0010: 8200 f662 c13a 1388 2166 4322 080a f400  ...b...fFC"....
  0x0020: 8010 0809 c534 0000 0101 080a 33bc 7052  ....4.....3.p0
  0x0030: 66a7 008c  f...
15:43:11.128930 IP (tos 0x2,ECT(0), ttl 50, id 4058, offset 0, flags [DF], proto TCP (6), length 53)
  130.208.246.98.complex-main > 192.168.1.137.49466: Flags [P,I], cksum 0xe131 (correct), seq 172, ack 1, win 510, options [nop,nop,TS val 1722224017 ecr 867987538], length 1
  0x0000: 4502 0035 0fd2 4000 3206 f0da 8208 f662  E..5..@.2....b
  0x0010: c0a8 0189 1388 c13a 088d f400 2166 4322  .....fFC"....
  0x0020: 0018 01fe e131 0000 0101 080a 66a7 0091  ....1.....f
  0x0030: 33bc 7052 0a 3.p0
15:43:11.128939 IP (tos 0x0, ttl 64, id 0, offset 0, flags [DF], proto TCP (6), length 52)
  192.168.1.137.49466 > 130.208.246.98.complex-main: Flags [I], cksum 0xe528 (correct), ack 2, win 2057, options [nop,nop,TS val 867987544 ecr 1722224017], length 0
  0x0000: 4500 0834 0000 4000 4005 f15f c0a8 0189  E..4..@.@....
  0x0010: 8200 f662 c13a 1388 2166 4322 088d f401  ...b...fFC"....
  0x0020: 8010 0809 c528 0000 0101 080a 33bc 7058  ....f.....3.pX
  0x0030: 66a7 0091  f...
15:43:23.413093 IP (tos 0x2,ECT(0), ttl 64, id 0, offset 0, flags [DF], proto TCP (6), length 95)
  192.168.1.137.49466 > 130.208.246.98.complex-main: Flags [P,I], cksum 0x284c (correct), seq 1444, ack 2, win 2057, options [nop,nop,TS val 867999828 ecr 1722224017], length 43
  0x0000: 4502 0035 0fd2 4000 3206 f0da 8208 f662  E..5..@.2....b
  0x0010: 8200 f662 c13a 1388 2166 4322 088d f401  ...b...fFC"....
  0x0020: 0018 0809 284c 0000 0101 080a 33bc a054  ....f.....3..T
  0x0030: 66a7 0091 5205 6d65 6d62 6572 2074 0f20  f...Remember.to.
  0x0040: c368 6563 6020 606f 7220 6060 7477 6f72  check.for.network
  0x0050: 6020 636f 606e 6563 7469 7669 7479 0a  r.network.connec
  0x0060:  tivity.
15:43:23.417154 IP (tos 0x2,ECT(0), ttl 50, id 4051, offset 0, flags [DF], proto TCP (6), length 52)
  130.208.246.98.complex-main > 192.168.1.137.49466: Flags [I], cksum 0xb0b0 (correct), ack 44, win 510, options [nop,nop,TS val 1722236306 ecr 867999828], length 0
  0x0000: 4502 0034 0fd3 4000 3206 f0da 8208 f662  E..4..@.2....b
  0x0010: c0a8 0189 1388 c13a 088d f401 2166 434d  .....fFCM
  0x0020: 8010 01fe 80b0 0000 0101 080a 66a7 3092  ....f...f.m.
  0x0030: 33bc a054 3..T
```

Question 1(c)

ncat -ln 5000:

Starts ncat in listening mode on port 5000, so the TSAM machine acts as a TCP server waiting for a connection.

ncat 130.208.246.98 5000:

Connects my computer to port 5000 on the TSAM server using TCP and lets me send data to it.

sudo tcpdump -vAX -i en0 'port 5000':

Captures packets on my Mac's en0 network interface involving port 5000. The options display detailed packet information and the packet contents in readable ASCII and hexadecimal form.

Question 1(d)

```
eric@erics-MacBook-Air:~$ sudo tcpdump -vAX -i en0 udp and port 5000
Password:
tcpdump: listening on en0, link-type EN10MB (Ethernet), snapshot length 524288 bytes
15:49:06.121622 IP (tos 0x0, ttl 64, id 52977, offset 0, flags [none], proto UDP (17), length 71)
  192.168.1.137.50237 > 130.208.246.98.complex-main: UDP, length 43
  0x0000: 4500 0047 cef1 0000 4011 7050 c0a8 0189  E..G....@.pP....
  0x0010: 82d0 f662 c43d 1388 0033 5fb0 5265 6d65  ...b.=...3_.Reme
  0x0020: 6d62 6572 2074 6f20 6368 6563 6b20 666f  mber.to.check.fo
  0x0030: 7220 6e65 7477 6f72 6b20 636f 6e6e 6563  r.network.connec
  0x0040: 7469 7669 7479 0a  tivity.
```

Question 1(e)

The TCP capture contained several packets because TCP first establishes a connection using a three-way handshake (SYN, SYN-ACK, ACK), and then uses acknowledgements when data is sent. In the UDP capture, there was no handshake; the message was sent directly in a UDP datagram. This difference occurs because TCP is connection-oriented, while UDP is connectionless.

Question 2

My scanner found these open UDP ports:

4013

4044

4081

I compared the result with Nmap using `sudo nmap -sU -p 4000-4100 130.208.246.98`.

Nmap also reported ports 4013, 4044, and 4081 as open. Nmap additionally reported port 4052 as open|filtered, meaning it could not determine whether the port was open or filtered. My scanner did not mark 4052 as open because it did not receive a response from that port.